

Cryogenic Leak Rate Dashboard — Verification Report

Date: 2026-05-08
Build version: 2.0.0
Status:  ALL CHECKS PASSED

1. Build Status

Check	Result
<code>python src/build_all.py</code>	 Success ({ "status": "ok", "builder_version": "2.0.0" })
Unit tests (7/7)	 All passed
OUTPUT_MANIFEST.json	 Generated with 125 files tracked

2. Required Output Formats

File	Path	Status
Navigation portal	<code>docs/index.html</code>	 Present & renders correctly
Interactive dashboard	<code>docs/dashboard.html</code>	 Present — 5 Plotly charts + scenario table
Calculation proofs	<code>docs/calculations.html</code>	 Present — formula proof, worked examples, validation checks, T/P matrix
Executive summary	<code>docs/executive_summary.html</code>	 Present — KPI cards, status badges, DMAIC note
RTM traceability	<code>docs/rtm_traceability.html</code>	 Present — RTM-047... RTM-067 mapping table
Handover (HTML)	<code>docs/handover.html</code>	 Present — full handover dossier with tables
Handover (PDF)	<code>docs/handover.pdf</code>	 Present (26 KB)

Supporting assets

- docs/assets/style.css ✓
- docs/assets/triage.js ✓
- docs/plots/ — 5 interactive Plotly HTML files ✓
- docs/source/ — 4 Markdown source files (handover, assumptions, developer_notes, rtm_traceability) ✓

3. Interactive Features Verified

Feature	Status	Notes
Sticky navigation bar	✓	Persistent across all pages (Portal, Dashboard, Calculations, Executive, RTM Traceability, Handover)
Navigation links	✓	All 6 nav links work correctly between pages
Status badges	✓	ACCEPT (green), REVIEW (orange), RISK (red) display correctly on Executive Summary and RTM pages
Plotly interactive charts	✓	All 5 plots are interactive (hover, zoom, pan) — log-log leak rate, temp/pressure effects, cost comparison, fleet sensitivity, reliability
Mode switching buttons	✓	Preview / Code-like / Print mode buttons present on all pages
Expand/Collapse all	✓	Buttons present and functional
Export PDF button	✓	Triggers browser print dialog with print-optimized CSS
DMAIC view notes	✓	Present on index, executive summary, dashboard, and calculations pages

4. Test Results

tests/test_build_outputs.py::test_required_outputs_exist	PASSED
tests/test_build_outputs.py::test_manifest_has_hashes	PASSED
tests/test_engineering.py::test_dimensional_chain	PASSED
tests/test_engineering.py::test_temperature_pressure_scaling	PASSED
tests/test_engineering.py::test_reference_rtm048_nm3_day_to_kg_year	PASSED
tests/test_engineering.py::test_baseline_scenario_total_range	PASSED
tests/test_engineering.py::test_unit_conversion_constant	PASSED

5. Key Computed Values (Validation Snapshot)

Metric	Value
Single valve @1e-8, 300K, 1 bar	0.000051 g/year
Baseline 410-valve scenario	34.278 kg/year
Uniform 410-valve @1e-5, 4K, 12 bar	18.686 kg/year
RTM-048 cap (1 Nm³/day)	65.20 kg/year

6. Triage System Requirements Compliance

Requirement	Status
Multi-audience routing (specialist/engineer/manager)	✓ Implemented via density levels
Engineering object classification (Requirement/Decision/Risk/Interface/Evidence/Validation)	✓ In all JSON data files
Status badge system (ACCEPT/REVIEW/RISK/TRACE/TODO/NEXT/DONE)	✓ CSS + HTML rendering
DMAIC view notes per output	✓ All pages include DMAIC blocks
Idempotent build system	✓ <code>write_text_if_changed</code> + SHA256 manifest
Output manifest with hashes	✓ <code>OUTPUT_MANIFEST.json</code> (125 files)
Print-optimized CSS	✓ <code>@media print</code> rules in <code>style.css</code>
Source traceability	✓ <code>source_anchors.json</code> + RTM page

7. Gaps / Issues

None identified. All required outputs are present, all navigation works, all interactive features function correctly, and all tests pass.